

Faiz Naufal Huda

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Summary

Computer Science undergraduate at IPB University (GPA: 3.82/4.00) with hands-on experience building production-grade systems end-to-end — from a full-stack academic document management system (118 tests, Clean Architecture) to an award-winning hackathon e-commerce platform and ML pipelines with CI/CD, Docker, and transformer fine-tuning. Selected participant in the Pijak × IBM SkillsBuild AI Engineer Cohort. Fluent in English (TOEFL ITP 607).

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, Bash
Frontend: React 18, Next.js, Vite, Tailwind CSS, Framer Motion
Backend & DB: FastAPI, SQLAlchemy 2.0, PostgreSQL, Supabase, JWT + bcrypt, Alembic
ML / Data: LightGBM, XGBoost, Scikit-learn, PyTorch, Hugging Face Transformers, TensorFlow/Keras, Pandas, NumPy
DevOps / Infra: Docker, GitHub Actions (CI/CD), MLflow, Prometheus, Grafana, Weights & Biases, Git
Other: ReportLab, PyPDF, qrcode, Pillow, Recharts

Experience

AI Engineer Cohort — Selected Participant

Jan 2026 – Present

Pijak × IBM SkillsBuild (Dicoding Indonesia)

- Completed structured learning path covering Python, supervised/unsupervised ML, deep learning (CNN & RNN), experiment tracking (MLflow), and production ML deployment via IBM SkillsBuild and Dicoding's industry curriculum.
- Built **PriorMail** as capstone — an email intelligence system with DistilBERT fine-tuning for 4-class priority classification; designed phishing detection module with security-focused adversarial evaluation protocol.
- Earned dual certificates from IBM SkillsBuild and Dicoding upon module completion.

Treasurer, Welfare & Internal Harmony Division

Jan 2025 – Dec 2025

HIMALKOM IPB University

- Managed annual divisional budget of ~IDR 15–20 million across 2–3 major programs, achieving 100% budget absorption with zero deficit — reflecting accurate planning and real-time expenditure tracking.
- Oversaw 20–30 financial transactions per program cycle; handled reimbursements and documentation with full accountability and transparent reporting.

Head of Funding & Consumption Division

Apr 2025 – Dec 2025

IT TODAY — HIMALKOM IPB University

- Managed IDR 33 million F&B consumption budget covering meals, snacks, and catering for 500+ participants and 90+ committee members across the full IT TODAY event series.
- Secured IDR 5 million in internal operational funding through entrepreneurial methods to support division activities.

Teaching Volunteer

Feb 2025 – Nov 2025

LPK Tepi Sawah — HIMALKOM IPB University

- Delivered 14 teaching sessions to 25+ children in Leuwiliang, Bogor over 9 months, covering literacy, English, mathematics, environmental awareness, and introductory programming through game-based activities.

- Served a dual role: committee organizer (coordination, logistics) and on-site volunteer (direct teaching and mentoring).

Projects

AQI Bangladesh — Air Quality Prediction (github.com/faizhuda/daming-kelompok-7) Apr – Jun 2026

- Independently built an end-to-end ML system predicting AQI across 30 cities in Bangladesh; trained LightGBM on 1M+ rows of multi-pollutant sensor data achieving MAE = 1.13, RMSE = 1.77, $R^2 = 0.9986$ — chosen over XGBoost for 3–10× faster training on large-scale tabular data.
- Engineered modular pipeline with centralized YAML config, lag features, rolling windows, cyclical time encoding, and walk-forward cross-validation (3-fold) for time-aware evaluation; experiment tracking via CSV logging.
- Implemented 40 unit tests across 6 modules (pytest + coverage); CI/CD via GitHub Actions with black/flake8 enforcement and pre-commit hooks; containerized with Docker for reproducible deployment.

AgritechJam — Agriinformatics 2025 (MyToko) (github.com/faizhuda/agritechjam-mytoko) Oct – Nov 2025

- **Placed 2nd** as full-stack developer in a 3-person team; built a complete customer + admin e-commerce platform using Next.js, TypeScript, Tailwind CSS, and Supabase (PostgreSQL) within hackathon timeframe.
- Implemented PostgreSQL stored procedures for atomic order cancellation with inventory rollback, eliminating race conditions on concurrent stock updates.
- Delivered real-time admin analytics dashboard (Recharts), PDF invoice generation, and full order lifecycle management (Pending → Paid → Shipped → Delivered).

AgriDesk — Academic Document Management System (github.com/faizhuda/ads-agridesk) Jan – Jun 2026

- Led full-stack development of a 4-person team project; solely responsible for the complete local implementation — backend in FastAPI + SQLAlchemy 2.0 + Alembic on PostgreSQL and frontend in React 18 + Vite + TailwindCSS v4 — built on Clean Architecture, Domain-Driven Design (DDD), and Finite State Machine document workflows with drag-and-drop PDF upload.
- Implemented SHA-256 document hashing with QR code verification enabling third-party authentication without system login; PDF generation (ReportLab + PyPDF) with coordinate-based digital signature stamping.
- Achieved 100% test coverage on core services across 118 passing tests; JWT + bcrypt authentication; in-memory SQLite for isolated unit testing.

Education

IPB University Jul 2023 – Oct 2027 (Expected)
Bachelor of Computer Science
GPA: 3.82 / 4.00

Certifications

- **AI Engineer Cohort** — IBM SkillsBuild + Dicoding Indonesia Jan 2026 – Present
- **Associate Data Scientist** — BNSP (Badan Nasional Sertifikasi Profesi) Jun 2026
- **Belajar Fundamental Deep Learning** — Dicoding Indonesia May 2026
- **Belajar Machine Learning untuk Pemula** — Dicoding Indonesia Mar 2026
- **Python for Data Science and ML Essential Training (Part 1 & 2)** — LinkedIn Learning Feb 2026
- **TOEFL ITP** — ETS (Score: 607) Nov 2023